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Article

Performance of smoking duration–based lung cancer screening eligibility criteria: a comparative modeling study

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Abstract

Background: Replacing the pack-year criterion for lung cancer screening with smoking duration has been proposed to reduce racial disparities in eligibility. However, no studies have evaluated the potential long-term benefits and harms of duration-based criteria. We conducted a comparative modeling study to evaluate the effectiveness of duration-based eligibility vs that from current guidelines and other alternative criteria.

Methods: We used 3 Cancer Intervention and Surveillance Modeling Network models to evaluate the performance of duration-based screening with various smoking duration thresholds (15-40 years in 5-year increments) for the 1960 and 1970 US birth cohorts. We first examined age-specific eligibility patterns of duration-based strategies vs pack-year and risk-based criteria. We then evaluated the performance of different strategies, comparing the resulting number of screens, lung cancer deaths averted, life-years gained, false-positive screens, and overdiagnosed cases. We compared the strategies' efficiency using lung cancer deaths averted and life-years gained per screen and the benefit-to-harm ratios using lung cancer deaths averted and life-years gained per overdiagnosed case.

Results: Risk-based criteria resulted in the most lung cancer deaths averted and life-years gained but also in more overdiagnosed cases. Duration-based strategies with a 35-year cutoff achieved comparable lung cancer deaths averted and life-years gained to current US guidelines and resulted in similar false-positive and overdiagnosed cases per screen. Duration-based scenarios with a 20-year cutoff required substantially more screenings but yielded only modest additional lung cancer deaths averted and life-years gained, resulting in lower benefits per screen than current guidelines.

Conclusion: Duration-based screening may be as efficient as current US guidelines. Given their potential to reduce disparities in eligibility shown in recent studies and simpler implementation, duration-based criteria warrant consideration.

Introduction

Lung cancer is the leading cause of cancer-related mortality in the United States.[1] Lung cancer screening via low-dose computed tomography (CT) is effective in reducing lung cancer mortality by detecting cancer at an earlier and more treatable stage.[2,3] Low-dose CT lung cancer screening is recommended by various guideline organizations. The US Preventive Services Task Force (USPSTF) currently recommends annual screening for individuals aged 50-80 years who have smoked 20 pack-years or more and currently smoke or quit within 15 years.[4,5] The American Cancer Society (ACS) recommends lung cancer screening for those aged between 50 and 80 years with over 20 pack-years smoking history but with no years-since-quit restriction.[6]

Recent updates to lung cancer screening guidelines aimed to better identify high-risk individuals and improve equity in eligibility. The USPSTF reduced pack-year requirements in 2021 to partially ameliorate racial disparities, while ACS further expanded the eligibility by removing the years-since-quit criterion in 2023 to include individuals who formerly smoked but are still at elevated risk of lung cancer.[4-7] Meanwhile, debate about relying on pack-years—a composite of smoking duration and intensity—to determine eligibility has been increasing. Although duration and intensity are independent lung cancer risk factors,[8-11] risk increases more sharply with duration (fourth or fifth power) than with intensity (second power).[10] Compared with pack-years, smoking duration is less prone to recall bias, more

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readily available in electronic health record systems, and applicable to other combustible tobacco products.[9] Prior studies have also suggested that incorporating smoking duration into eligibility criteria may reduce racial disparities.[12-15] For example, Potter et al.[13] found that replacing the 20-pack-year criterion in USPSTF guidelines with 20 years of smoking duration improved screening sensitivity and reduced racial disparities in eligibility.

Given these potential benefits, the latest National Comprehensive Cancer Network Guidelines (Version 1.2025) added individuals aged 50 years and older with 20 years or more of cigarette smoking as a high-risk group (evidence and consensus category 2B).[16] Nevertheless, the panel noted that such simplified criteria could lead to overscreening and overtesting.[16] To date, no studies have yet examined the impact of duration-based criteria on the general US population. To address this gap, we use 3 Cancer Intervention and Surveillance Modeling Network (CISNET) lung cancer models, which supported the USPSTF and ACS screening guidelines,[4,6] to project the long-term benefits and harms of duration-based criteria and compare their performance with current pack-year or risk-based screening strategies for the US population.

Methods

We used 3 CISNET lung cancer natural history models, which informed the USPSTF 2021 and ACS 2023 lung cancer screening guidelines.[4,6] These include the Lung Cancer Interventions Decision Simulation model (previously known as the BCCRI-LungCan or MichiganLung model), the Microsimulation Screening Analysis-Lung Model from Erasmus University, and the Stanford Lung Cancer Outcomes Simulation model. All models simulate lung cancer natural history based on age, sex, and smoking history, sharing similar lung cancer risk models but differing in histology and stage distributions, survival models, and assumptions about sojourn time and screening performance.[17] Further details are available in the Supplement (Table S1) and prior literature.[4,6]

Study population

Individual smoking histories were simulated using the CISNET Smoking History Generator.[18-20] Consistent with previous work,[4,6,21-23] we simulated 1 million men and 1 million women from the US 1960 birth cohort. We also simulated individuals from the 1970 birth cohort to evaluate the performance of different screening eligibility criteria for a younger cohort. The CISNET lung models then simulated individual-level lung cancer–related outcomes from age 45 to 90 years under alternative screening criteria.

Study outcomes

We calculated the age-specific percentage eligibility for each screening scenario. Results from the 3 CISNET models were aggregated to derive population-level mean outcomes across the models. For each eligibility scenario, screening-related benefits included lung cancer deaths averted and life-years gained, whereas harms included false-positive screens and overdiagnosed cases. We evaluated resource needs by the number of screens. We assessed screening efficiency using lung cancer deaths averted or life-years gained per screen and the number of individuals who needed to be screened (NNS) to avoid 1 lung cancer death. We also estimated benefit-to-harm ratios, calculated as lung cancer deaths averted or life-years gained per overdiagnosed case. Results are presented as counts per 100 000 individuals alive at age 45 years rather than per the screened population so that the results are comparable across eligibility scenarios. Model-specific results are presented in the Supplement (Tables S10-S21).

Screening scenarios

We compared the performance of 3 sets of screening eligibility criteria, totaling 28 scenarios, relative to no screening. We assumed perfect screening uptake and adherence to focus solely on the impact of varying screening eligibility criteria on lung cancer outcomes.[24] However, we explored the impact of imperfect adherence in sensitivity analyses. All scenarios assumed screening occurring between ages 50 and 80 years, except the USPSTF 2013 criteria (55-80 years).

Duration-based strategies (12 scenarios) included thresholds of 15-40 years of smoking duration (5-year increments) with and without years-since-quit constraint: duration-threshold 15 years since quit and duration-threshold no years since quit. Duration-threshold no-years-since-quit strategies are also referred to as duration-only strategies.

Pack-year–based strategies (8 scenarios) included TF-2013 (USPSTF 2013 criteria; 30 pack-years or more and no more than 15 years since quitting), TF-2021 (USPSTF 2021 criteria; 20 pack-years or more and no more than 15 years since quitting), ACS-2023 (20 pack-years or more and no years since quitting limit), and 5 strategies (PackYear-10-years since quittingthreshold) with 10 pack-years of smoking criterion and varying years since quitting thresholds (15, 20, 25, 30, and none).

Risk-based strategies (8 scenarios) used the Prostate, Lung, Colorectal, and Ovarian Cancer Screening Trial (PLCO) lung cancer risk prediction model (PLCOm2012)[25] with risk thresholds of 0.5%, 0.8%, 1%, 1.2%, 1.5%, 1.7%, 1.8%, and 2% (PLCO-risk threshold).

Sensitivity analysis

We conducted a sensitivity analysis for all scenarios, excluding individuals with limited life expectancy from screening. Specifically, individuals were eligible to initiate or continue screening only if they had at least 5 years of remaining life expectancy, as determined by the Smoking History Generator–simulated age at death from nonlung cancer causes. Moreover, we varied the probability of receiving screening from the baseline 100% to 20% or 30% to evaluate how realistic uptake and adherence rates affect the benefits and harms of screening.

Results

Screening eligibility for the study population

Figure 1 shows the estimated age-specific percentage eligible for screening for the US 1960 birth cohort under different scenarios. For duration-only strategies with duration cutoffs of at least 35 years, eligibility rose sharply with age until 55-60 years, followed by a plateau. In contrast, scenarios with cutoffs less than 35 years had relatively flat eligibility curves across age, with little to no increase beyond age 50 years. For duration-based strategies with a 15YSQ constraint and duration cutoffs of at least 35 years, eligibility initially followed the same increasing pattern as the duration-only strategies until age 55-60 years but then declined rapidly instead of plateauing. With cutoffs less than 35 years, eligibility decreased with age. In comparison, pack-year–based strategies generally showed declining eligibility with age, except for ACS-2023 and PackYear-10 with no years-since-quit constraint, which maintained relatively constant eligibility. Risk-

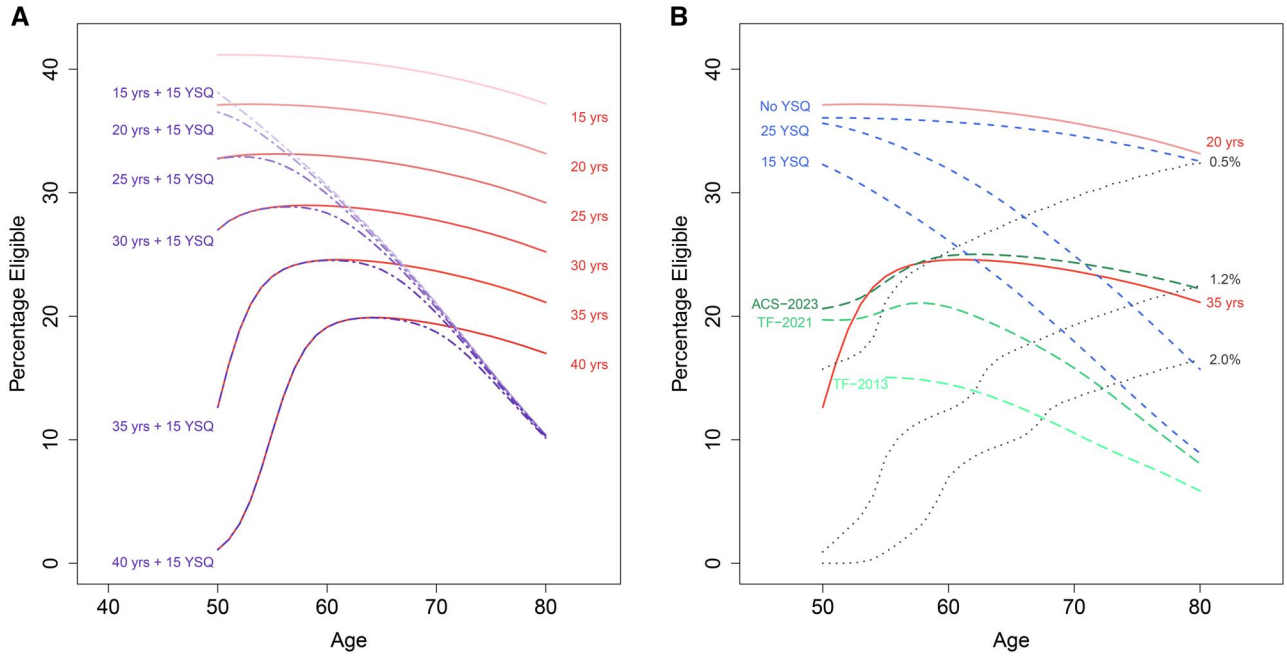


Figure 1. Percentage eligible for screening by age for duration-only **(red solid lines)** and duration-based with 15YSQ constraint **(purple dash-dotted lines)** strategies **(A)** and duration-only **(red solid lines)**, packyear-10 **(blue short-dashed lines)**, and PLCOm2012 risk-based **(black dotted lines)** scenarios **(B)** for the 1960 birth cohort. **Green long-dashed lines** represent the percentage of screening eligibility under 3 US screening guidelines (TF-2013, TF-2021, and ACS-2023). Age-specific percentage eligible was calculated as the number of individuals eligible for screening over the number of individuals at risk at a given age. Abbreviations: ACS = American Cancer Society; TF = US Preventive Services Task Force; YSQ = years since quit.

based strategies displayed a continuous increase in eligibility with age for all risk thresholds. These patterns were consistent for the 1970 birth cohort (Figure S1).

Screening outcomes

Figure 2 shows the estimated screening benefits and harms per 100 000 population for each screening scenario by the number of low-dose CT screens for the 1960 birth cohort. In general, risk-based scenarios had higher lung cancer deaths averted and life-years gained than other strategies with similar numbers of screens but also resulted in a higher number of overdiagnoses. Compared with TF-2021 and ACS-2023 guidelines, the duration-35-15YSQ and duration-35-noYSQ strategies resulted in comparable lung cancer deaths averted and overdiagnosed cases at a similar number of screens, but the associated life-years gained were lower. Compared to duration-only strategies, duration-based strategies with 15YSQ constraint performed similarly when the duration cutoff was greater than or equal to 35 years but gained fewer benefits when the duration cutoff was less than 35 years and had fewer overdiagnosed cases overall. PackYear-10 strategies with varying years since quit had fewer lung cancer deaths averted but similar life-years gained compared with duration-only scenarios with a similar number of screens.

Figure 2, A and B, shows the model-estimated screening benefits—lung cancer deaths averted and life-years gained per 100 000 population—by the number of low-dose CT screens for the 1960 birth cohort. Across all strategies, benefits increased with the number of screens but at a diminishing rate. Adding a 15YSQ constraint to duration-based strategies would considerably reduce the number of screens and overall benefits, especially when the duration cutoff was below 35 years but had less impact for duration cutoffs of 35 years or more. Duration-based strategies with 35- or 40-year cutoffs achieved equal or more lung cancer deaths averted and life-years gained per screen than the ACS-2023 and TF-2021 guidelines, respectively. In contrast, duration-based strategies with cutoffs of 30 years or less had fewer benefits per screen than current guidelines. PackYear-10 strategies, despite having fewer lung cancer deaths averted, achieved life-years gained comparable with duration-only strategies with a similar number of screens. Risk-based strategies generally achieved the highest lung cancer deaths averted and life-years gained for a given number of screens.

Figure 2, C and D, shows model-projected screening-related harms, including false-positives and overdiagnosed cases per 100 000 population by the number of low-dose CT screens for the 1960 birth cohort. The number of false-positives per 100 000 increased linearly with the number of low-dose CT screens across all scenarios. Overdiagnosis, however, varied more substantially across eligibility criteria. Given the same number of screens, risk-based strategies led to the highest number of overdiagnosed cases, followed by pack-year–based or duration-only strategies, and then duration-based strategies with a 15YSQ constraint. Duration-based strategies with 35 years or less cutoffs resulted, in general, in fewer overdiagnosed cases per screen than the TF-2021 and ACS-2023 guidelines.

Tables 1 and 2 summarize the model-projected benefits, harms, and efficiency of all screening strategies for the 1960 birth cohort. Duration-35-15YSQ had a higher number of screens than TF-2021 (520 463 vs 453 074; 14.9% more) with a similar percentage eligible (24.6% vs 24.4%). It also achieved slightly more lung cancer deaths averted (591 vs 574 per 100 000; 3.0% higher) and marginally higher life-years gained (8191 vs 8161; 0.4% higher) but resulted in more overdiagnosed cases (99 vs 93 per 100 000; 6.5% increase) (Table 1). Duration-35-15YSQ had a slightly less favorable benefit-to-harm trade-off than TF-2021, with 6 vs 6.2 lung cancer deaths averted per overdiagnosis (3.3% lower) and

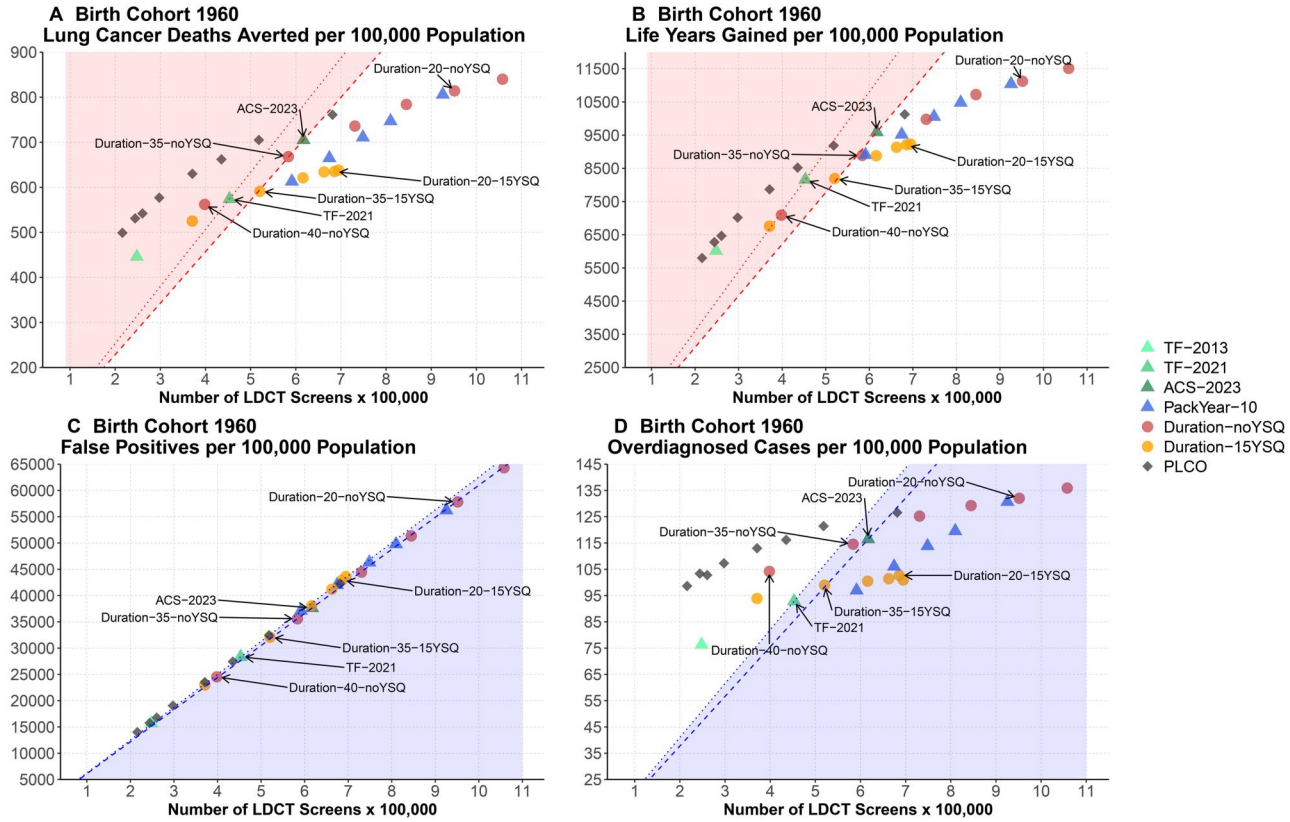


Figure 2. Number of LDCT screens vs LCDA and life-years gained **(A and B)** and number of false-positive and overdiagnosed cases **(C and D)** averaged across 3 microsimulation models for the 1960 birth cohorts. **Red dots** from left to right represent duration-only scenarios with smoking duration over 40, 35, 30, 25, 20, and 15 years; **orange dots** from left to right are duration-based scenarios with a 15YSQ constraint and duration over 40, 35, 30, 25, 20, and 15 years; **gray diamonds** from left to right represent risk-based scenarios with thresholds being 2%, 1.8%, 1.7%, 1.5%, 1.2%, 1%, 0.8%, and 0.5%; **blue triangles** from left to right are packyear-10 strategies with years-since-quit thresholds of 15, 20, 25, 30, and no years-since-quit constraint. **Red lines** represent the slope of outcomes (benefits or harms) per LDCT screen (**dotted** = TF-2021 and **dashed** = ACS-2023). The **red shaded area** highlights strategies that yield equal or greater benefits per LDCT screen than at least 1 of the current guidelines (TF-2021 or ACS-2023). The **blue shaded area** highlights strategies with equal or fewer harms per LDCT screen than at least 1 of the current guidelines. Abbreviations: ACS = American Cancer Society; LCDA = lung cancer deaths averted; LDCT = low-dose computed tomography; TF = US Preventive Services Task Force; YSQ = years since quit; PLCO denotes PLCOm2012 risk-based scenarios.

82.8 vs 87.9 life-years gained per overdiagnosis (6.2% lower) (Table 2). NNS was the same for both strategies.

Duration-35-noYSQ had fewer screens than ACS-2023 (583 675 vs 617 134; 5.4% fewer) with a similar percentage eligible (24.6% vs 25.2%) (Table 1). It led to slightly fewer lung cancer deaths averted (668 vs 705; 5.2% fewer) than ACS-2023 and lower life-years gained (8889 vs 9592; 7.3% lower). The benefit-to-harm trade-off for duration-35-noYSQ compared with ACS-2023 was 5.8 vs 6.1 lung cancer deaths averted per overdiagnosis (4.9% lower) and 77.6 vs 82.3 life-years gained per overdiagnosis (5.7% lower) (Table 2). The NNS was nearly identical (37 vs 36).

Replacing the 20 pack-years with 20 years of duration in the TF-2021 guideline (duration-20-15YSQ) would result in 10.6% more lung cancer deaths averted and 12.9% more life-years gained but with a considerable increase in the number of screens (685 576 vs 453 074; 51.3% increase) (Table 1). The duration-20-noYSQ strategy would result in 41.8% more lung cancer deaths averted and 36.3% more life-years saved compared with the TF-2021 scenario but require more than double the number of screens (951 612). Both duration-20 scenarios have fewer lung cancer deaths averted and life-years gained per screen and higher NNS compared with the TF-2021 and ACS-2023 strategies (Table 2). The lung cancer deaths averted per overdiagnosis were 6.2 for both duration-20 scenarios, which are similar to those of current guidelines. However, their life-years gained per overdiagnosis of 89.7 (duration-20-15YSQ) and 84.3 (duration-20-noYSQ) were higher than those from the TF-2021 and ACS-2023 strategies, respectively.

The number of screens, lung cancer deaths averted, life-years gained, false-positives, and overdiagnoses were lower for the 1970 birth cohort than those from the 1960 birth cohort (Figure S2; Tables S2 and S3); however, the relative performance of screening scenarios was similar for both cohorts. NNS was higher, and the benefit-to-harm ratio was lower for the 1970 birth cohort, reflecting its lower smoking rates vs the 1960 cohort.

Sensitivity analysis

With the 5-year life expectancy constraint, the estimated lung cancer deaths averted and life-years gained would reduce across strategies for the 1960 birth cohort by 3.3% and 0.2% on average, respectively, while false-positives were reduced by 5.1% and overdiagnosis by 69.9%. This led to improved benefit-to-overdiagnosis ratios compared with those without the life expectancy constraint (Figure 3; Tables S4 and S5). Results for the 1970 birth cohort show similar patterns (Figure S3; Tables S6 and S7). With 20% uptake and adherence, the number of screens would decrease by 80%, lung cancer deaths averted and life-years gained by 69%, false-positives by 73%, and overdiagnosis by 65%. At 30% uptake/

Table 1. Benefits and harms of all screening strategies under sensitivity analysis for the 1960 birth cohort, average model results per 100 000 population[a]

Scenario	% eligible	Low-dose computed tomography screens	Mean low-dose computed tomography screens per person screened	Screen-detected lung cancer cases	Lung cancer deaths averted	Life-years gained	False-positive cases	Mean false-positive results per person screened	Overdiagnosed cases	Overdiagnosis: % of screen-detected lung cancer cases
NoScreen	0.0	0	0.0	0	0	0	0	0.0	0	0.0
PLCO-2.0%	16.8	215 977	12.8	1491	499	5799	14 011	0.8	99	6.6
PLCO-1.8%	18.2	244 184	13.4	1566	531	6278	15 762	0.9	103	6.6
TF-2013	15.6	247 934	15.9	1301	446	6018	15 710	1.0	76	5.9
PLCO-1.7%	18.9	260 456	13.8	1607	542	6466	16 774	0.9	103	6.4
PLCO-1.5%	20.5	297 644	14.5	1691	577	7016	19 054	0.9	107	6.3
Duration-40-15YSQ	19.7	370 792	18.8	1530	525	6756	22 997	1.2	94	6.1
PLCO-1.2%	23.3	370 948	15.9	1826	630	7867	23 523	1.0	113	6.2
Duration-40-noYSQ	19.7	398 211	20.3	1652	562	7091	24 526	1.2	104	6.3
PLCO-1%	25.6	435 322	17.0	1916	662	8520	27 422	1.1	116	6.1
TF-2021	24.4	453 074	18.6	1631	574	8161	28 338	1.2	93	5.7
PLCO-0.8%	28.4	518 204	18.3	2016	705	9187	32 442	1.1	121	6.0
Duration-35-15YSQ	24.6	520 463	21.2	1703	591	8191	32 076	1.3	99	5.8
Duration-35-noYSQ	24.6	583 675	23.8	1919	668	8889	35 579	1.4	115	6.0
PackYear-10-15	31.9	591 339	18.5	1732	613	8908	37 122	1.2	97	5.6
Duration-30-15YSQ	28.9	615 906	21.3	1770	621	8879	38 046	1.3	100	5.7
ACS-2023	25.2	617 134	24.5	2007	705	9592	37 635	1.5	116	5.8
Duration-25-15YSQ	32.8	662 500	20.2	1795	634	9135	41 207	1.3	101	5.6
PackYear-10-20	33.9	674 385	19.9	1890	665	9519	42 016	1.2	106	5.6
PLCO-0.5%	33.7	681 322	20.2	2169	761	10 127	42 281	1.3	127	5.8
Duration-20-15YSQ	36.1	685 576	19.0	1806	635	9213	42 894	1.2	103	5.7
Duration-15-15YSQ	37.6	694 720	18.5	1808	638	9231	43 620	1.2	101	5.6
Duration-30-noYSQ	28.9	730 750	25.3	2096	736	9978	44 451	1.5	125	6.0
PackYear-10-25	35.1	748 578	21.3	2011	711	10 056	46 315	1.3	114	5.7
PackYear-10-30	35.5	809 831	22.8	2104	747	10 481	49 790	1.4	120	5.7
Duration-25-noYSQ	32.9	844 867	25.7	2215	784	10 722	51 350	1.6	129	5.8
PackYear-10-noYSQ	35.5	925 507	26.1	2272	806	11 043	56 226	1.6	131	5.8
Duration-20-noYSQ	36.6	951 612	26.0	2300	814	11 125	57 823	1.6	132	5.7
Duration-15-noYSQ	40.4	1 057 848	26.2	2371	840	11 511	64 308	1.6	136	5.7

Abbreviations: ACS = American Cancer Society; TF = US Preventive Services Task Force; YSQ = years since quit; PLCO denotes the PLCOm2012 risk-based scenarios

[a] Individuals were followed up from age 45 to 90 years. Count results were rescaled to per 100 000 population alive at age 45 years.

Table 2. Efficiency metrics of all screening strategies under sensitivity analysis for the 1960 birth cohort, average model results per 100 000 population[a]

Scenario	Low-dose computed tomography screens	NNS[b]	Lung cancer deaths averted per screens ($\times 10^{-4}$)	Life-years gained per screens ($\times 10^{-2}$)	Lung cancer deaths averted per overdiagnosis	Life-years gained per overdiagnosis
NoScreen	0	0	0.0	0.0	0.0	0.0
PLCO-2.0%	215 977	34	23.1	2.7	5.1	58.8
PLCO-1.8%	244 184	34	21.8	2.6	5.1	60.7
TF-2013	247 934	35	18.0	2.4	5.8	78.8
PLCO-1.7%	260 456	35	20.8	2.5	5.3	62.9
PLCO-1.5%	297 644	35	19.4	2.4	5.4	65.4
Duration-40-15YSQ	370 792	38	14.1	1.8	5.6	72.0
PLCO-1.2%	370 948	37	17.0	2.1	5.6	69.6
Duration-40-noYSQ	398 211	35	14.1	1.8	5.4	68.1
PLCO-1%	435 322	39	15.2	2.0	5.7	73.3
TF-2021	453 074	42	12.7	1.8	6.2	87.9
PLCO-0.8%	518 204	40	13.6	1.8	5.8	75.7
Duration-35-15YSQ	520 463	42	11.4	1.6	6.0	82.8
Duration-35-noYSQ	583 675	37	11.4	1.5	5.8	77.6
PackYear-10-15	591 339	52	10.4	1.5	6.3	91.8
Duration-30-15YSQ	615 906	47	10.1	1.4	6.2	88.4
ACS-2023	617 134	36	11.4	1.6	6.1	82.3
Duration-25-15YSQ	662 500	52	9.6	1.4	6.3	90.1
PackYear-10-20	674 385	51	9.9	1.4	6.3	89.7
PLCO-0.5%	681 322	44	11.2	1.5	6.0	80.0
Duration-20-15YSQ	685 576	57	9.3	1.3	6.2	89.7
Duration-15-15YSQ	694 720	59	9.2	1.3	6.3	91.5
Duration-30-noYSQ	730 750	39	10.1	1.4	5.9	79.7
PackYear-10-25	748 578	49	9.5	1.3	6.2	88.3
PackYear-10–30	809 831	48	9.2	1.3	6.2	87.7
Duration-25-noYSQ	844 867	42	9.3	1.3	6.1	83.0
PackYear-10-noYSQ	925 507	44	8.7	1.2	6.2	84.5
Duration-20-noYSQ	951 612	45	8.6	1.2	6.2	84.3
Duration-15-noYSQ	1 057 848	48	7.9	1.1	6.2	84.7

Abbreviations: ACS = American Cancer Society; NNS = Number Needed to Screen; TF = US Preventive Services Task Force; YSQ = years since quit; PLCO denotes the PLCOm2012 risk-based scenarios.

[a] Individuals were followed up from age 45 to 90 years. Count results were rescaled to per 100 000 population alive at age 45 years.

[b] $NNS = \frac{Number\ of\ Individuals\ Ever\ Received\ Screenings}{Lung\ Cancer\ Deaths\ Averted} = \frac{100{,}000 \times Percent\ Eligible}{Lung\ Cancer\ Deaths\ Averted\ per\ 100{,}000}$, where the percent eligible and lung cancer deaths averted per 100 000 could be found in Table 1.

adherence, the number of screens would decrease by 70%, lung cancer deaths averted and life-years gained by 56%, false-positives by 64%, and overdiagnosis by 51%. These reductions were consistent across scenarios and cohorts (Tables S8 and S9). The relative ranking of eligibility strategies remained similar to that observed under 100% uptake/adherence (Figures S4 and S5).

Discussion

We evaluated and compared the benefits and harms of lung cancer screening eligibility criteria based on smoking duration against pack-year–based and risk-based strategies. Overall, risk-based strategies result in the most lung cancer deaths averted and life-years gained as well as in the highest number of overdiagnosed cases. Duration-based scenarios with longer duration cutoffs (≥35 years) maintained comparable efficiency with the current ACS and USPSTF guidelines in terms of lung cancer deaths averted and life-years gained per screen. In contrast, duration-based strategies with lower cutoffs (≤20 years) extended eligibility to younger individuals, increasing the total number of screens and reducing efficiency. However, their lower overdiagnosis rates preserved favorable benefit-to-harm trade-offs like those from current guidelines. Importantly, none of the duration-based criteria simultaneously achieved greater benefits and fewer harms than current recommendations.

Our study complements previous studies by projecting long-term outcomes of duration-based strategies. Potter et al.[13] showed that duration-20-15YSQ greatly increased screening sensitivity among lung cancer patients in the Southern Community Cohort Study compared with the USPSTF 2021 and eliminated racial disparities in eligibility between Black and White individuals. However, according to our analysis, the duration-20-15YSQ strategy, while resulting in an 11% increase in lung cancer deaths averted and a 13% increase in life-years gained compared with the USPSTF 2021, would require more than 50% more screenings. The NNS for duration-20-15YSQ is 57 vs 42 for the USPSTF 2021. Similarly, duration-20-noYSQ results in more than a 100% increase in the number of screens but a less than 50% increase in lung cancer deaths averted and life-years gained compared with USPSTF 2021. However, because these 2 strategies include more individuals at younger ages for screening, the overdiagnosis rates are lower, leading to comparable benefit-to-harm ratios as current guidelines.

Given that the additional screening volume required by duration-20 strategies yields only modest incremental benefits, such approaches would be difficult to justify, particularly in health-care systems with limited resources. Nonetheless, although these strategies might not be ideal as sole criteria, duration-20-noYSQ may serve as a rule-in step to identify high-risk individuals, as most individuals aged between 50 and 80 years who smoked had a 20-year smoking history (National

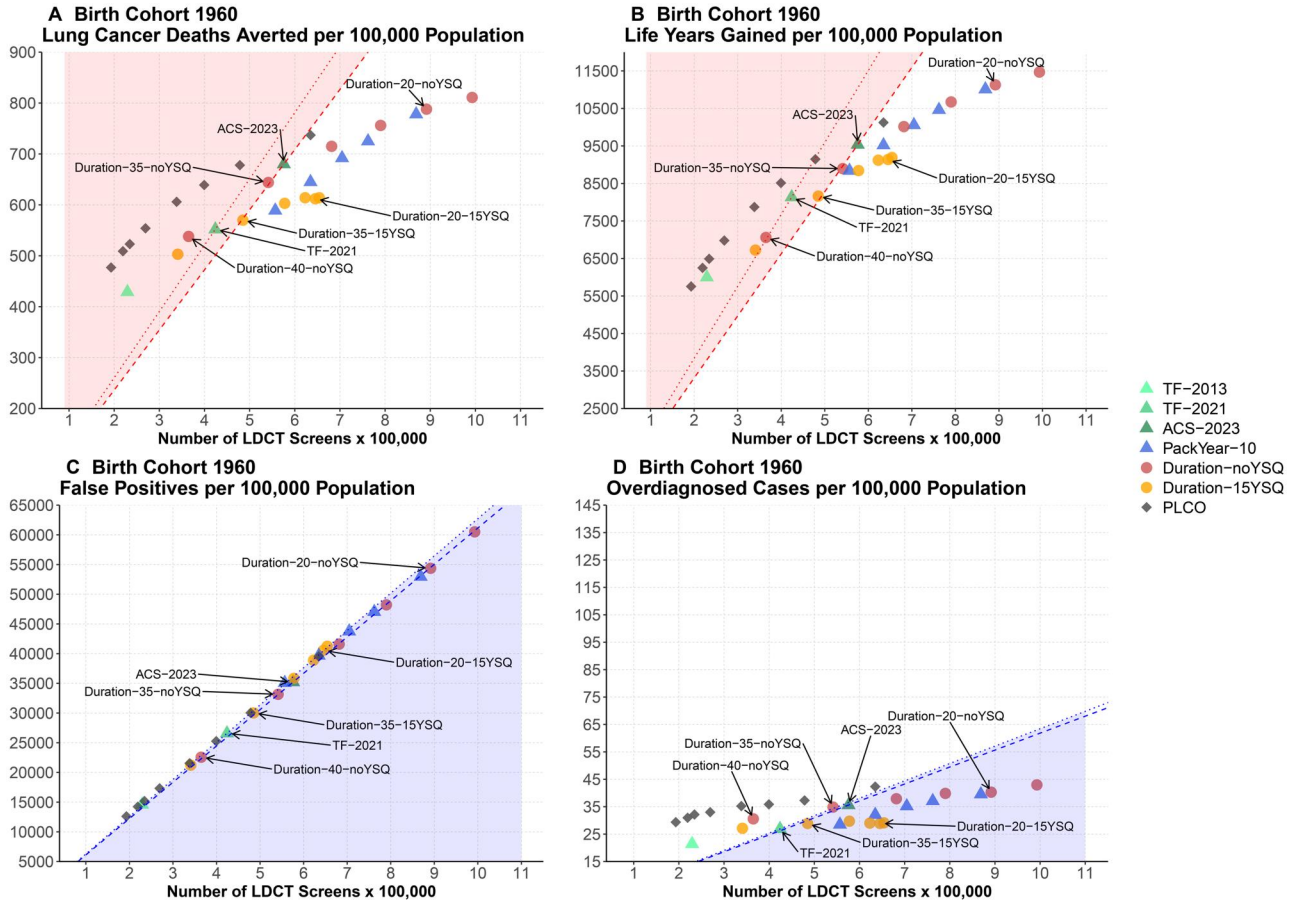


Figure 3. Results of sensitivity analysis excluding individuals with less than 5 years of life expectancy: Number of LDCT screens vs LCDA and life-years gained **(A and B)** and number of false-positive and overdiagnosed cases **(C and D)** across 3 microsimulation models for the 1960 birth cohort. Plot legends are identical to Figure 2. Abbreviations: ACS = American Cancer Society; LCDA = lung cancer deaths averted; LDCT = low-dose computed tomography; TF = US Preventive Services Task Force; YSQ = years since quit; PLCO denotes PLCOm2012 risk-based scenarios.

Health Interview Survey 2020 data, Table S22). Further assessment, such as risk calculation, could then be used to determine eligibility as a second step before performing a cost-intensive series of CT scans. This 2-step process would reduce the burden on primary care, as staff would only need to determine smoking duration and then refer potential individuals to dedicated screening programs or cancer centers for comprehensive risk assessment. For example, in British Columbia, Canada, primary care providers help identify patients who might benefit from lung cancer screening by referring individuals aged 55-74 years with 20 or more years of smoking to the provincial Lung Screening Program.[26] Program navigators then assess their eligibility using the PLCOm2012 model. Individuals with a 6-year lung cancer risk greater than 1.5% are then referred for screening. Additionally, the rule-in first step could help identify more current smokers who would benefit from smoking cessation services but are not yet eligible for lung cancer screening, which in turn could lead to considerable gains in health outcomes.

Alternatively, duration-based strategies with a 35-year duration criterion could serve as alternatives or augmentations to current US screening guidelines. They demonstrate comparable efficiencies and benefit-to-harm trade-offs to the current guidelines while potentially simplifying the identification of eligible individuals—an important advantage given ongoing challenges with incomplete smoking information and pack-year calculation in clinical settings.[9,27-32] Such criteria may help improve screening uptake in the United States, which remains below 20%.[33,34] Consistent with our findings, Kearney et al.[14] also found that screening those who have smoked for at least 40 years would identify individuals with potentially high benefits from screening. Nonetheless, using longer duration thresholds (eg, 35-40 years) would delay screening to older adults, reducing life-years gained and increasing overdiagnosis. Given the higher comorbidity burden and frailty in older individuals with high smoking exposure,[35-39] the net benefit of screening may be reduced. Thus, evaluating patients' overall health status remains essential when considering screening.

Our analysis relied on 3 well-validated CISNET lung cancer natural history models. These models have been used to inform the decision making of lung cancer screening guidelines in the United States.[4,6,40] Even though the absolute numbers of predictions differed slightly between models, the overall patterns and conclusions from these 3 independently developed models were consistent, increasing the validity of our findings. Furthermore, the relative performance of screening criteria was consistent for both the 1960 and 1970 birth cohorts, suggesting a potential transferability of findings to younger cohorts.

Our study also has limitations. First, although duration-based strategies have been shown to reduce racial disparities in screening, our microsimulation models, based on the general US population, did not compare the performance of alternative strategies for specific racial and ethnic groups. Further efforts are needed

to evaluate whether duration-based strategies may outperform the current guidelines for specific race and ethnicity groups using race-specific lung cancer natural history models.[41] Second, comorbidities or frailty were not explicitly modeled, and our analysis relied on an idealized assumption of perfect life-expectancy assessment. Although imperfect, this approach illustrates that limiting screening to individuals with adequate life expectancy could preserve benefits while reducing harms. It also underscores the need for practical, clinically implementable life-expectancy models that incorporate comorbidity and frailty. Third, this study focused solely on cigarette use, though other tobacco product use, such as e-cigarettes, is increasing in the United States.[42] Emerging evidence suggests that dual use of e-cigarettes and cigarettes may elevate lung cancer risk compared with exclusive cigarette use,[43] while polytobacco use has been linked to higher mortality risks.[44] Duration-based strategies may better capture exposure for more than 1 tobacco product, so future studies are needed to evaluate their performance in the evolving tobacco landscape. Relatedly, our analysis only considered age, sex, and smoking history and did not incorporate other risk factors such as race and ethnicity or chronic obstructive pulmonary disease, which are included in the full PLCOm2012 model; however, prior studies have shown that models based primarily on age and smoking variables perform similarly in calibration and discrimination,[21,45] suggesting limited impact on estimating the overall population-level screening benefits and harms. Finally, although our sensitivity analysis incorporated more realistic uptake and adherence levels (20% and 30%), these rates were applied uniformly and did not vary by prior screening results, individual characteristics, or eligibility strategy. In practice, uptake and adherence may differ across strategy types. For instance, duration-based criteria could facilitate higher uptake because of simpler implementation, while adherence to subsequent rounds may be comparable across strategies. More research is needed to characterize how eligibility criteria might influence real-world screening behaviors, and future modeling studies should incorporate such data as they become available.

In conclusion, our study provides comprehensive evidence on the benefits, harms, and resource needs of duration-based strategies relative to current and alternative criteria. Our findings show that strategies using a 35-year or longer duration threshold can achieve comparable efficiency and resource needs as the ACS and USPSTF recommendations. In contrast, strategies with a cutoff of no more than 20 years are less efficient in benefits per screen, with a substantial increase in screening volume. These findings can help policy makers and guideline committees evaluate trade-offs and prioritize criteria based on their specific goals and resource contexts.

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Author contributions

Pianpian Cao (Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing—original draft, Writing—review & editing), Koen de Nijs (Conceptualization, Formal analysis, Investigation, Methodology, Writing—review & editing), Justin Lee (Formal analysis, Writing—review & editing), Jihyoun Jeon (Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Writing—original draft, Writing—review & editing), Harry J. De Koning (Conceptualization, Funding acquisition, Methodology, Writing—review & editing), Sylvia K. Plevritis (Conceptualization, Funding acquisition, Methodology, Writing—review & editing), Kevin Ten Haaf (Conceptualization, Data curation, Formal analysis, Methodology, Writing—review & editing), Summer S. Han (Conceptualization, Formal analysis, Investigation, Methodology, Writing—review & editing), and Rafael Meza (Conceptualization, Funding acquisition, Investigation, Methodology, Supervision, Writing—original draft, Writing—review & editing)

Supplementary material

Supplementary material is available at *Journal of the National Cancer Institute* online.

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Conflict of interest

The authors have no conflict of interest to disclose.

Data availability

The data underlying this article, including smoking prevalence, initiation, cessation, and intensity rates from 1964 to 2100, are available in the CISNET Smoking Parameter repository, at us.cisnetsmokingparameters.org. The Smoking History Generator simulation code used in this study can be made available to academic researchers upon approval by the CISNET Lung Steering Committee. Access will not be granted to tobacco industry employees or affiliates.

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